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### LIQUID EJECTING HEAD

#### Abstract

A liquid ejecting head includes a channel member having: a common channel extending in a first direction orthogonal to an up-down direction; and individual channels. Individual channel rows each including the individual channels aligned in the first direction are disposed side by side in a second direction orthogonal to the up-down direction and the first direction. A first pressure chamber and a second pressure chamber overlap in the second direction. A first nozzle included in the first individual channel row and a second nozzle included in the second individual channel row are located on one side in the second direction with respect to the common channel. The first nozzle is located closer to the common channel than the second nozzle is in the second direction, and the first nozzle overlaps with the second pressure chamber in the up-down direction.

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## Background/Summary

### REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priorities from Japanese Patent Application No. 2024-032167 filed on Mar. 4, 2024 and Japanese Patent Application No. 2024-032169 filed on Mar. 4, 2024. The entire contents of the priority applications are incorporated herein by reference.

### BACKGROUND ART

[0002] A known ink-jet recording head includes two ink pools (common channels) disposed in an overlapping manner in the up-down direction, and a plurality of individual channels each including a nozzle and a pressure chamber communicating with the nozzle. In a certain known ink-jet recording head, the plurality of individual channels includes a plurality of first individual channels each communicating with an upper ink pool of the two ink pools and a plurality of second individual channels each communicating with a lower ink pool of the two ink pools, and these first individual channels and second individual channels are located in a partially overlapping manner in the up-down direction, and a plurality of nozzles are located at high density.

### SUMMARY

[0003] In the above-described ink-jet recording head, the plurality of ink pools are disposed in the overlapping manner in the up-down direction, and pressure chambers each of which is included in the plurality of second individual channels are located between the ink pools. This increases the height of the ink-jet recording head. In a case where the height of each of the ink pools is reduced in order to reduce the height of the ink-jet recording head, the volume of each of the ink pools is reduced, resulting in a shortage in the supply of ink from each of the ink pools to the respective individual channels.

[0004] In this regard, an object of the present disclosure is to provide a liquid ejecting head which contributes to preventing the head from becoming large-sized while locating nozzles at high density and maintaining the volume of the common channels.

[0005] A liquid ejecting head according to an aspect of the present disclosure includes a channel member having: a common channel extending in a first direction orthogonal to an up-down direction, and a plurality of individual channels each including a nozzle and a pressure chamber communicating with the nozzle, each of the plurality of individual channels communicating with the common channel. A plurality of individual channel rows each including the plurality of individual channels aligned in the first direction are disposed side by side in a second direction orthogonal to the up-down direction and the first direction. A first pressure chamber and a second pressure chamber overlap in the second direction, the first pressure chamber being included in a first individual channel row among the plurality of individual channel rows and the second pressure chamber being included in a second individual channel row adjacent to the first individual channel row among the plurality of individual channel rows. A first nozzle included in the first individual channel row and a second nozzle included in the second individual channel row are located on one side in the second direction with respect to the common channel. The first nozzle is located closer to the common channel than the second nozzle is in the second direction, and the first nozzle overlaps with the second pressure chamber in the up-down direction.

[0006] According to the liquid ejecting head of the present disclosure, the first nozzle overlaps with the second pressure chamber in the up-down direction, and thus a part of the communicating channel communicating with the first pressure chamber and the first nozzle included in the first individual channel row also overlaps with the second pressure chamber in the up-down direction. This contributes to locating the plurality of individual channels connected to the same common channel at high density. Therefore, the liquid ejecting head of the present disclosure is capable of contributing to preventing the channel member from becoming large-sized, while locating the

plurality of nozzles at high density, maintaining length in the second direction (channel width) and height of the common channel, and securing the volume of the common channel.

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## Description

### BRIEF DESCRIPTION OF DRAWINGS

[0007] FIG. 1 is a plan view of a printer including a head 1.

[0008] FIG. 2 is a block diagram depicting the electrical configuration of the printer.

[0009] FIG. 3 is a plan view of the head 1.

[0010] FIG. 4 is an enlarged view of an area IV depicted in FIG. 3 on the upper surface of a channel member.

[0011] FIG. 5 is a cross-sectional view of the head 1 along a V-V line in FIG. 4.

[0012] FIG. 6 is a cross-sectional view of the head 1 along a VI-VI line in FIG. 4.

[0013] FIG. 7 is a vertical cross-sectional view of the main components of another head 1.

[0014] FIG. 8A is a vertical cross-sectional view of the main components of yet another head 1, and FIG. 8B is a plan view of a horizontal part 317A2A depicted in FIG. 8A.

[0015] FIG. 9 is a plan view of a head 401.

[0016] FIG. 10 is an enlarged view of an area X depicted in FIG. 9 on the upper surface of a channel member.

[0017] FIG. 11A is a cross-sectional view of the head 401 along a XIA-XIA line in FIG. 10, and FIG. 11B is a cross-sectional view of the head 401 along a XIB-XIB line in FIG. 10.

[0018] FIG. 12A is a cross-sectional view of the head 401 along a XIIA-XIIA line in FIG. 10, and FIG. 12B is a cross-sectional view of the head 401 along a XIIB-XIIB line in FIG. 10.

[0019] FIG. 13 is an enlarged view depicting the first to fourth individual channels, and two common channels depicted in FIG. 10.

[0020] FIG. 14A is a view depicting a state in which first to fourth nozzles are located in a case where the head 401 is inclined by a predetermined angle  $\theta$  with respect to a first direction in a plane orthogonal to the up-down direction, and FIG. 14B is a view depicting a state in which the first to fourth nozzles are located in a case where another head is inclined by the predetermined angle  $\theta$  with respect to the first direction in the plane orthogonal to the up-down direction.

### DESCRIPTION

#### First Embodiment

[0021] First, the overall configuration of a printer 100 including a head 1 according to a first embodiment of the present disclosure will be described, with reference to FIG. 1. Note that in the following description, a first direction D1 and a second direction D2 are horizontal directions and are orthogonal to an up-down direction D3. The up-down direction D3 in the present embodiment is along the vertical direction, but the up-down direction D3 may be an up-down direction crossing the vertical direction and the horizontal direction. The first direction D1 is orthogonal to the second direction D2.

#### Overall Configuration of Printer

[0022] The printer 100 includes a casing 100A, a head unit 1X, a platen 3, a conveyor 4, and a controller 5. The head unit 1X, the platen 3, the conveyor 4, and the controller 5 are disposed inside the casing 100A. The printer 100 further includes a button (not depicted in the drawings) disposed in the outer surface of the casing 100A.

[0023] The length of the head unit 1X in the first direction D1 is longer than the length of the head unit 1X in a conveyance direction along the second direction D2. The first direction D1 is a direction along the width of a sheet 9. The head unit 1X is fixed to the casing 100A. The type of the head unit 1X is a line system.

[0024] The head unit 1X includes four heads 1. The four heads 1 are disposed in a staggered

manner in the first direction D1. The length in the first direction D1 of each of the four heads 1 is longer than the length in the second direction D2 of each of the four heads 1.

[0025] The platen 3 is a plate along the plane orthogonal to the up-down direction D3, and is disposed below the head unit 1X. The sheet 9 is supported on the upper surface of the platen 3.

[0026] The conveyor 4 has two roller pairs A4 and 4B disposed, with the platen 3 interposed therebetween in the second direction D2. In a case where a conveying motor 4C is driven by control of the controller 5, the two roller pairs A4 and 4B rotate while holding the sheet 9, and the sheet 9 is conveyed in the conveyance direction along the second direction D2.

[0027] As depicted in FIG. 2, the controller 5 includes a CPU 51, a ROM 52, and a RAM 53. The CPU 51 executes various kinds of control in accordance with a program and data stored in the ROM 52 and/or the RAM 53, based on data input from an external apparatus or the above-described button. The external apparatus is, for example, a personal computer (PC).

[0028] The ROM 52 stores the program and data to be used in a case where the CPU 51 performs the various kinds of control. The RAM 53 temporarily stores data to be used in a case where the CPU 51 executes the program.

#### Head 1

[0029] As depicted in FIG. 3, each of the four heads 1 has a channel member 21 and an actuator member 22. Both the channel member 21 and the actuator member 22 have a rectangular shape in which the length thereof in the first direction D1 is longer than the length thereof in the second direction D2 in the plane orthogonal to the up-down direction D3.

[0030] As depicted in FIG. 3, six supply ports 111 and six return ports 112 are open in an upper surface (surface) 21A of the channel member 21. The six supply ports 111 are located at one end in the first direction D1 of the channel member 21. The six return ports 112 are located at the other end in the first direction D1 of the channel member 21. Each of the six supply ports 111 and the six return ports 112 is connected to an ink tank via a tube. The channel member 21 has six common channels 12, a plurality of individual channels 13, and six damper chambers 19.

[0031] The six common channels 12 are disposed side by side in the second direction D2 and each extend in the first direction D1. Each of the six supply ports 111 is connected to one end in the first direction D1 of one of the six common channels 12. Each of the six return ports 112 is connected to the other end in the first direction D1 of one of the six common channels 12. Each of the six common channels 12 communicates with the ink tank via a corresponding supply port 111 among the six supply ports 111 and a corresponding return port 112 among the six return ports 112, and communicates with the plurality of individual channels 13.

[0032] Each of the six damper chambers 19 is located below a corresponding common channel 12 among the six common channels 12. The six damper chambers 19 are also disposed side by side in the second direction D2 and extend in the first direction D1.

[0033] As depicted in FIGS. 4 to 6, each of the plurality of individual channels 13 includes one of a plurality of nozzles 15, one of a plurality of pressure chambers 16, one of a plurality of communicating channels 17, and one of a plurality of connecting channel 18. One end of the communicating channel 17 communicates with the nozzle 15, and the other end of the communicating channel 17 communicates with the pressure chamber 16. One end of the connecting channel 18 communicates with the common channel 12, and the other end of the connecting channel 18 communicates with the pressure chamber 16. The communicating channel 17 communicates with one end of the pressure chamber 16, and the connecting channel 18 communicates with the other end of the pressure chamber 16.

[0034] As depicted in FIG. 5, the channel member 21 includes eleven plates 121 to 131. Note that the channel member 21 may be constructed of eleven or more plates or ten or less plates. The plurality of pressure chambers 16 are formed in the plate 121 (the uppermost layer) among the eleven plates 121 to 131, and the plurality of nozzles 15 are formed in the plate 131 (the lowermost layer) among the eleven plates 121 to 131.

[0035] The plurality of pressure chambers **16** are open in the upper surface (upper surface **21A**) of the plate **121**, and the plurality of nozzles **15** are open in the lower surface of the plate **131**. The opening of each of the nozzles **15** is circular, and the opening of each of the pressure chambers **16** is substantially rectangular which is elongated in the second direction D2. In other words, the length (width) in the first direction D1 of the pressure chamber **16** is shorter than a length in the second direction D2 of the pressure chamber **16**. The nozzle **15** has a shape tapered downward, as depicted in FIGS. 5 and 6.

[0036] Each of the common channels **12** is constructed of two holes formed in the two plates **127**, **128**, respectively, and connected to each other. Each of the common channels **12** overlaps with all the pressure chambers **16** which communicate with each of the common channels **12** in the up-down direction D3.

[0037] Each of the six damper chambers **19** is constructed by closing a hole formed in a plate **130** with the two plates **129**, **131**. Parts, of the plate **129**, each of which is sandwiched by one of the damper chambers **19** and one of the common channels **12** corresponding thereto, function as dampers **129A** which absorb the change in pressure of the ink inside the respective common channels **12**. In other words, even in a case where pressure generated in a certain pressure chamber **16** during ink ejection from a certain nozzle **15** corresponding to the certain pressure chamber **16** is propagated to the common channel **12**, the damper **129A** elastically deforms so as to attenuate the pressure, thereby preventing the phenomenon in which the pressure generated in the certain pressure chamber **16** is propagated to another pressure chamber **16** (so-called crosstalk).

[0038] As depicted in FIG. 3, the plurality of individual channels **13** are aligned in a first direction D1 so as to form **24** individual channel rows **14**. These **24** individual channel rows **14** are disposed side by side in the second direction D2. Four rows of the **24** individual channel rows **14** correspond to one of the six common channels **12**. For example, as depicted in FIGS. 4 to 6, four first individual channel row **14** which are a first individual channel row **14A** to a fourth individual channel row **14D** include first individual channels **13A**, second individual channels **13B**, third individual channels **13C** and fourth individual channels **13D**, respectively. The first individual channels **13A** to the fourth individual channels **13D** communicate with the same common channel **12** among the six common channels **12**. Therefore, it can be considered that four rows of the **24** individual channel rows **14** correspond to one of the six common channels **12**.

[0039] As depicted in FIG. 4, among the first individual channel rows **14A** to the fourth individual channel rows **14D** corresponding to each of the six common channels **12**, each of first nozzles **15A** included in the first individual channel row **14A** and each of second nozzles **15B** included in the second individual channel row **14B** are located on one side in the second direction D2 (the right side in FIG. 4) with respect to a corresponding common channel **12** among the six common channels **12**. Among the first individual channel rows **14A** to fourth individual channel rows **14D** corresponding to each of the six common channels **12**, each of third nozzles **15C** included in the third individual channel row **14C** and each of fourth nozzles **15D** included in the fourth individual channel row **14D** are located on the other side in the second direction D2 (the left side in FIG. 4) with respect to a corresponding common channel **12** among the six common channels **12**.

[0040] The first individual channel **13A** included in the first individual channel row **14A** and the third individual channel **13C** included in the third individual channel row **14C** have the same channel structure including the channel shape and size. More specifically, the first individual channel **13A** and the third individual channel **13C** are located point-symmetrically with respect to a middle point of a line segment connecting the first nozzle **15A** and the third nozzle **15C** in the plane orthogonal to the up-down direction D3.

[0041] The second individual channel **13B** included in the second individual channel row **14B** and the fourth individual channel **13D** included in the fourth individual channel row **14D** also have the same channel structure including the channel shape and size. More specifically, the second individual channel **13B** and the fourth individual channel **13D** are located point-symmetrically with

respect to a middle point of a line segment connecting the second nozzle **15B** and the fourth nozzle **15D** in the plane orthogonal to the up-down direction **D3**.

[0042] The detailed configuration of each of the first individual channel **13A** and the second individual channel **13B** will be described below.

[0043] As depicted in FIG. 5, each of the first individual channels **13A** includes a first nozzle **15A**, a first pressure chamber **16A**, a communicating channel **17A**, and a connecting channel **18A**. As depicted in FIG. 4, the first pressure chamber **16A** overlaps, in the second direction **D2**, with a second pressure chamber **16B** included in a second individual channel **13B** adjacent to the first individual channel **13A**. The connecting channel **18A** corresponds to a “first connecting channel” of the present disclosure.

[0044] The communicating channel **17A** has a first up-down channel **17A1** and a first communicating channel **17A2**. The first up-down channel **17A1** extends upward from the first nozzle **15A** along the up-down direction **D3**. As depicted in FIG. 4, the first up-down channel **17A1** overlaps with the second pressure chamber **16B** included in the second individual channel **13B** in the up-down direction **D3**. Further, the first up-down channel **17A1** is constructed of five holes formed, respectively, in the five plates **126** to **130** which are connected to one another, and has a diameter greater than the diameter of the first nozzle **15A**.

[0045] Furthermore, the first nozzle **15A** also overlaps with the second pressure chamber **16B** in the up-down direction **D3**. In the second direction **D2**, the first nozzle **15A** is located closer to the corresponding common channel **12** than the second nozzle **15B** included in the second individual channel **13B** is.

[0046] The first communicating channel **17A2** has a horizontal part **17A2A** connected to the upper end of the first up-down channel **17A1** and a vertical part **17A2B** connected to the first pressure chamber **16A**. The horizontal part **17A2A** is a hole formed in the plate **125**, and extends from the upper end of the first up-down channel **17A1** in a fourth direction **D4** parallel to the second direction **D2**. As depicted in FIG. 4, the horizontal part **17A2A** in the present embodiment has the length in the first direction **D1** which is approximately the same as the length in the first direction **D1** of the first up-down channel **17A1**. The horizontal part **17A2A** corresponds to a “first extending part” of the present disclosure.

[0047] As depicted in FIG. 5, the vertical part **17A2B** is constructed of three holes formed, respectively, in the three plates **122** to **124** and connected to one another. The vertical part **17A2B** extends upward toward the first pressure chamber **16A** from an end, of the horizontal part **17A2A**, which is close to the first pressure chamber **16A**. In this manner, the first communicating channel **17A2** is constructed of the four holes formed, respectively, in the four plates **122** to **125** and connected to one another.

[0048] As depicted in FIG. 5, the connecting channel **18A** has a first part **18A1** and a second part **18A2**. The first part **18A1** is constructed of three holes formed, respectively, in the three plates **124** to **126** and connected to one another, and is connected to the upper end of the common channel **12**.

[0049] The second part **18A2** has a horizontal part **18A2A** connected to the upper end of the first part **18A1** and a vertical part **18A2B** connected to the first pressure chamber **16A**. The horizontal part **18A2A** is a hole formed in the plate **123**, and extends from the upper end of the first part **18A1** toward the first pressure chamber **16A** along a sixth direction **D6** (see FIG. 4) which crosses the first direction **D1** and the second direction **D2** and is orthogonal to the up-down direction **D3**.

Further, as depicted in FIG. 4, the horizontal part **18A2A** has a throttle part **18A2C** which throttles a flow amount of the liquid. The throttle part **18A2C** is formed in a central part in the sixth direction **D6** of the horizontal part **18A2A**, and has a channel width (length in the direction orthogonal to the sixth direction **D6**) smaller than a channel width of the first part **18A1**. In the present embodiment, although both ends in the sixth direction **D6** of the horizontal part **18A2A** have a channel width greater than the channel width of the throttle part **18A2C**, the horizontal part **18A2A** may have a channel width which is constant over the entirety of the length of the horizontal

part **18A2A**. The horizontal part **18A2A** corresponds to a “third extending part” of the present disclosure.

[0050] As depicted in FIG. 5, the vertical part **18A2B** is a hole formed in the plate **122**, and extends upward toward the first pressure chamber **16A**, from an end, of the horizontal part **18A2A**, which is close to the first pressure chamber **16A**. In this manner, the second part **18A2** is constructed of the two holes formed, respectively, in the two plates **122**, **123** and connected to each other.

[0051] As depicted in FIG. 6, the second individual channel **13B** includes a second nozzle **15B**, a second pressure chamber **16B**, a communicating channel **17B**, and a connecting channel **18B**. The communicating channel **17B** corresponds to a “second up-down channel” of the present disclosure, and the connecting channel **18B** corresponds to a “second connecting channel” of the present disclosure.

[0052] The communicating channel **17B** extends upward along the up-down direction **D3** from the second nozzle **15B**. The communicating channel **17B** is constructed of nine holes formed, respectively, in the nine plates **122** to **130** and connected to one another, and has a diameter greater than the diameter of the second nozzle **15B**.

[0053] The channel shape and size of the communicating channel **17B** are adjusted so that the channel resistance and inertance of the communicating channel **17B** are the same as the channel resistance and inertance of the communicating channel **17A**. The inertance of a channel is expressed as  $\rho L/S$  [kg/m.sup.4], in a case where  $S$  [m.sup.2] is the cross-sectional area of the channel,  $L$  [m] is the length of the channel, and  $\rho$  [kg/m.sup.3] is the density of the ink inside the channel.

[0054] As depicted in FIG. 6, the connecting channel **18B** has a first part **18B1** and a second part **18B2**. The first part **18B1** is constructed of three holes formed, respectively, in the three plates **124** to **126** and connected to one another, and is connected to the upper end of the common channel **12**. A connecting position at which the connecting channel **18B** is connected to the common channel **12** is separated, in the second direction **D2**, from a connecting position at which the connecting channel **18A** is connected to the common channel **12**.

[0055] The second part **18B2** has a horizontal part **18B2A** connected to the upper end of the first part **18B1** and a vertical part **18B2B** connected to the second pressure chamber **16B**. The horizontal part **18B2A** is a hole formed in the plate **123**, and extends from the upper end of the first part **18B1** toward the second pressure chamber **16B** along a fifth direction **D5** parallel to the second direction **D2**, as depicted in FIG. 4. Further, as depicted in FIG. 4, the horizontal part **18B2A** has a throttle part **18B2C** which throttles the flow amount of the liquid. The horizontal part **18B2A** corresponds to a “second extending part” of the present disclosure.

[0056] The horizontal part **18B2A** of the connecting channel **18B** is located at a different height from the horizontal part **17A2A** of the first communicating channel **17A2**. With this, the horizontal part **17A2A** and the horizontal part **18B2A** can be disposed so as to overlap in the up-down direction **D3**, without interfering with each other, as depicted in FIG. 4. This eliminates the need to greatly separate the first individual channel **13A** and the second individual channel **13B** in the horizontal direction, allowing the first individual channels **13A** and the second individual channels **13B** to be disposed at high density. Owing to this configuration, the size of the channel member **21** is less likely to be great in the direction orthogonal to the up-down direction **D3**.

[0057] As depicted in FIG. 6, the vertical part **18B2B** is a hole formed in the plate **122**, and extends upward toward the second pressure chamber **16B** from an end, of the horizontal part **18B2A**, which is close to the second pressure chamber **16B**. In this manner, the second part **18B2** is constructed of the two holes formed, respectively, in the two plates **122**, **123** and connected to each other.

[0058] In the present embodiment, although the connecting channel **18A** and the connecting channel **18B** have the different connecting positions and inclinations with respect to pressure chambers **16**, with which the connecting channel **18A** and the connecting channel **18B** communicate, respectively, in the plane orthogonal to the up-down direction **D3**, the connecting

channel **18A** and the connecting channel **18B** have the same channel length and the same channel width. In other words, the connecting channel **18A** and the connecting channel **18B** have the same channel shape and size.

[0059] The first pressure chamber **16A** and the connecting channel **18A** included in the first individual channel **13A** and the second pressure chamber **16B** and the connecting channel **18B** included in the second individual channel **13B** have the same channel shape and size, and also have the same channel resistance and the same inertance. Further, the communicating channel **17A** included in the first individual channel **13A** and the communicating channel **17B** included in the second individual channel **13B** have the different channel shapes and sizes but have the same channel resistance and inertance. Accordingly, the first individual channel **13A** and the second individual channel **13B** have the same channel resistance and inertance.

[0060] The configuration of the third individual channel **13C** and the configuration of the fourth individual channel **13D** are similar, respectively, to the configuration of the first individual channel **13A** and the configuration of the second individual channel **13B** as described above, and thus the third individual channel **13C** and the fourth individual channel **13D** will not be described in detail. As depicted in FIG. 6, the third individual channel **13C** includes a third nozzle **15C**, a third pressure chamber **16C**, a communicating channel **17C**, a third up-down channel **17C1**, a third communicating channel **17C2**, a horizontal part **17C2A**, a vertical part **17C2B**, a connecting channel **18C**, a first part **18C1**, a second part **18C2**, a horizontal part **18C2A**, a vertical part **18C2B**, and a throttle part **18C2C**. The connecting channel **18C** corresponds to a “third connecting channel” of the present disclosure.

[0061] As depicted in FIG. 5, the fourth individual channel **13D** includes a fourth nozzle **15D**, a fourth pressure chamber **16D**, a communicating channel **17D**, a connecting channel **18D**, a first part **18D1**, a second part **18D2**, a horizontal part **18D2A**, a vertical part **18D2B**, and a throttle part **18D2C**.

[0062] As depicted in FIG. 4, the third pressure chamber **16C** overlaps with the fourth pressure chamber **16D** in the second direction **D2**. More specifically, the first pressure chamber **16A** to the fourth pressure chamber **16D** included, respectively, in the first individual channel **13A** to the fourth individual channel **13D** overlap with one another in the second direction **D2**. Further, the third nozzle **15C** also overlaps with the fourth pressure chamber **16D** in the up-down direction **D3**. In the second direction **D2**, the third nozzle **15C** is located closer to the corresponding common channel **12** than the fourth nozzle **15D** is. The connecting channel **18A** corresponds to the “first connecting channel” of the present disclosure.

[0063] As depicted in FIG. 4, the horizontal part **18C2A** of the connecting channel **18C** extends in the sixth direction **D6** and is located side by side, in the first direction **D1**, with the horizontal part **18A2A** included in the connecting channel **18A**. This allows the first individual channel **13A** and the third individual channel **13C** to be located close to each other in the second direction **D2**. The horizontal part **18C2A** corresponds to a “fourth horizontal part” of the present disclosure.

[0064] In each of the individual channel rows **14**, as depicted in FIG. 4, the plurality of nozzles **15** are aligned in the first direction **D1** at a predetermined pitch **P**. Two nozzles **15** included, respectively, in two individual channels **13** adjacent to each other in the second direction **D2** are located apart in the first direction **D1** by a distance corresponding to  $1/24$  of the pitch **P**. With this, in a case where the recording resolution at the pitch **P** is 100 dpi, recording resolution of 2400 dpi is realized by all the nozzles **15**.

[0065] The ink in the ink tank is supplied to each of the six common channels **12** via one of the six supply ports **111** corresponding thereto by driving a pump **10** depicted in FIG. 2 under the control of the controller **5**, and then the ink is distributed from the six common channels **12** to the plurality of individual channels **13**.

[0066] In each of the plurality of individual channels **13**, the volume of the pressure chamber **16** is reduced by driving an actuator part **35** which will be described later, and pressure is applied to the



ink in the pressure chamber **16**, thereby causing the ink to pass through the communicating channel **17** and to be ejected as an ink droplet from the nozzle **15**.

[0067] The ink supplied from each of the six supply ports **111** moves in one of the six common channels **12** corresponding thereto from the one end to the other end in the first direction **D1**, and reaches one of the six return ports **112** corresponding thereto. The ink which reaches each of the six return ports **112** is returned to the ink tank via the tube.

[0068] As depicted in FIGS. **3**, **5**, and **6**, the actuator member **22** is fixed to the upper surface **21A** of the channel member **21**. The actuator member **22** includes a vibration plate **31** made of metal, a piezoelectric layer **32**, and a plurality of individual electrodes **33**.

[0069] Parts, of the actuator member **22**, each of which overlaps with one of the plurality of pressure chambers **16** corresponding thereto in the up-down direction **D3** function as actuator parts **35**. Each of the actuator parts **35** is independently deformable in accordance with the potential applied to one of the plurality of individual electrodes **33** corresponding thereto.

[0070] Each of the actuator parts **35** is a thin-film piezoelectric element. The thin-film piezoelectric element is a so-called micro electro mechanical system (MEMS). The actuator parts **35** are formed by sequentially depositing a thin film which becomes the piezoelectric layer **32** and a thin film which becomes the plurality of individual electrodes **33**, on the upper surface of the vibration plate **31**.

[0071] The vibration plate **31** is disposed on the upper surface **21A** of the channel member **21** so as to cover the plurality of pressure chambers **16**. The piezoelectric layer **32** is disposed on the upper surface of the vibration plate **31**. Each of the plurality of individual electrodes **33** is disposed on the upper surface of the piezoelectric layer **32** so as to overlap with one of the pressure chambers **16** corresponding thereto in an up-down direction **D3**.

[0072] The vibration plate **31** and the plurality of individual electrodes **33** are electrically connected to a driver IC **6** (FIG. **2**). The driver IC **6** maintains the potential of the vibration plate **31** at the ground potential, whereas the driver IC **6** changes the potential of each of the plurality of individual electrodes **33**. The vibration plate **31** functions as a common electrode which is common to the plurality of actuator parts **35**.

[0073] The driver IC **6** generates a drive pulse signal based on a control signal from the controller **5**, and supplies the drive pulse signal to each of the plurality of individual electrodes **33**. The drive pulse signal changes the potential of each of the plurality of individual electrodes **33** between a predetermined driving potential and the ground potential. In this manner, the actuator part **35** is driven, and the pressure is applied to the ink in the pressure chamber **16**, causing an ink droplet to pass through the communicating channel **17** and to be ejected from the nozzle **15**.

[0074] As described above, according to the head **401** of the present embodiment, owing to the first nozzle **15A** overlapping with the second pressure chamber **16B** in the up-down direction **D3**, a part of the communicating channel **17A**, which communicates with the first pressure chamber **16A** and the first nozzle **15A** included in the first individual channel row **14A** also overlaps with the second pressure chamber **16B** in the up-down direction **D3**, and the individual channels **13A**, **13B** connected to the same common channel **12** can be located at high density. In other words, even in a case where the common channels are not located to overlap each other vertically, the individual channels **13A**, **13B** can be located at high density while maintaining the length in the second direction **D2** (channel width) of the common channel **12**. Therefore, the channel member **12** can be prevented from becoming large-sized while locating the plurality of nozzles **15** at high density, maintaining the length in the second direction **D2** and height of each of the common channels **12**, and securing the volume of each of the common channels **12**.

[0075] The head **1** includes the actuator member **22** having the plurality of actuator parts **35** each corresponding to one of the plurality of pressure chambers **16**. In this case, the head **1** is less susceptible to the influence of non-uniformity during the manufacture of the actuator member **22** itself than in a case where the head **1** includes a plurality of actuator members each of which has an

actuator part **35** corresponding to one of the pressure chambers **16**. Therefore, in the head **1** of the present embodiment, the ink ejection characteristic is less likely to vary among the plurality of nozzles **15**.

[0076] The first individual channel **13A** and the second individual channel **13B** have the same channel resistance. As a result, the channel resistances of the first individual channel **13A** to the fourth individual channel **13D** are also the same. Owing to this, the ink flows easily from each of the common channels **12** to any one of the plurality of individual channels **13**. Therefore, even in a case where the same pressure is applied to the ink in the plurality of pressure chambers **16**, the ink ejection characteristic is less likely to vary among the plurality of nozzles **15**.

[0077] The first individual channel **13A** and the second individual channel **13B** have the same inertance. Owing to this, the first individual channel **13A** to the fourth individual channel **13D** also have the same inertance. This reduces non-uniformity in the acoustic length (AL) which is a round-trip propagation time of a pressure wave in each of the plurality of individual channels **13**. As a result, the ink ejection characteristic is less likely to vary among the plurality of nozzles **15**.

[0078] Further, the connecting channels **18A** and **18B** have the same channel length and the same channel width. Owing to this, the first individual channel **13A** and the second individual channel **13B** are more likely to have the same channel resistance or the same inertance.

[0079] Furthermore, the first individual channel **13A** and the third individual channel **13C**, which communicate with the same common channel **12**, have the same channel configuration, and the second individual channel **13B** and the fourth individual channel **13D**, which communicate with the same common channel **12**, have the same channel configuration. Owing to this, the ink ejection characteristic is less likely to vary among the plurality of nozzles **15**, in a similar manner as described above.

[0080] The first communicating channel **17A2** included in the first individual channel **13A** has the horizontal part **17A2A** extending in the fourth direction **D4**. Owing to this, even in a case where the first up-down channel **17A1** is shorter than the communicating channel **17B**, the adjustment so as to make the channel resistance and/or inertance of the first individual channel **13A** and the second individual channel **13B** the same can be easily performed by the horizontal part, of the first communicating channel **17A2A**, which extends in the fourth direction **D4**.

[0081] The connecting position at which the connecting channel **18B** connects to the common channel **12** is separated, in the second direction **D2**, from the connecting position at which the connecting channel **18A** connects to the common channel **12**. In a presumed case where these two connecting positions are not separated in the second direction **D2** but are located side by side in the first direction **D1**, the ink at the same position in the common channel **12** flows to the individual channel **13** in the second direction **D2**. Due to this, ink each at positions separated from the connecting position with respect to the common channel **12** in the second direction **D2** stagnates without flowing to the individual channel **13**. However, in the present embodiment, since these two connecting positions are separated in the second direction **D2**, the ink flows from at least two positions in the second direction **D2** of the common channel **12** to the individual channel **13**. Therefore, the stagnation of the ink is less likely to occur in the common channel **12**.

#### Modifications

[0082] As a first modification, the first individual channel **13A** may include a communicating channel **217A** depicted in FIG. 7, instead of the above-described communicating channel **17A**. The communicating channel **217A** in the first modification has a first up-down channel **217A1** and a first communicating channel **217A2**, as depicted in FIG. 7. The first up-down channel **217A1** extends upward from the first nozzle **15A** along the up-down direction **D3**. The first up-down channel **217A1** overlaps with the second pressure chamber **16B** included in the second individual channel **13B** in the up-down direction **D3**. Further, the first up-down channel **217A1** is constructed of four holes formed, respectively, in the four plates **127** to **130** and connected to one another, and has a diameter greater than the diameter of the first nozzle **15A**.

[0083] The first communicating channel **217A2** communicates with the first pressure chamber **16A** and the first up-down channel **217A1**. The first communicating channel **217A2** extends, without bending, at a location below the second pressure chamber **16B** in a third direction **D7** which crosses the second direction **D2** and the up-down direction **D3**. The first communicating channel **217A2** overlaps, in the up-down direction **D3**, with a common channel **12** which is one of the six communication channels **12** and with which the first communicating channel **217A2** communicates. Moreover, the first communicating channel **217A2** is constructed of five holes formed, respectively, in the five plates **122** to **126** and connected to one another. More specifically, the first communicating channel **217A2** is constructed of the five holes formed, respectively, in the five plates **122** to **126** and connected to one another in a stepped manner.

[0084] With the first communicating channel **217A2** configured in such a manner, the channel resistance in the first communicating channel **217A2** becomes smaller than the channel resistance in a channel bent at, for example, 90 degrees between the first pressure chamber **16A** and the first up-down channel **217A1**. Owing to this, the ink can easily flow from the first pressure chamber **16A** toward the first nozzle **15A**.

[0085] Further, the channel shape and size of the communicating channel **217A** are adjusted so that the communicating channel **217A** has the channel resistance and inertance which are the same as the channel resistance and inertance of the above-described communicating channel **17A**, thereby achieving the effect similar to the above-described effect.

[0086] Furthermore, the upper surface of the common channel **12** has a protruded part **212A** which protrudes into the common channel **12** in an area overlapping with the first communicating channel **217A2**. The protruded part **212A** of the first modification is constructed of an upper side wall part, of the common channel **12**, which is close to the first nozzle **15A**. The upper side wall part is constructed of a hole which constructs a part, of the common channel **12**, which is formed in the plate **127** and is formed to have a smaller hole diameter than a hole diameter of a hole constructing another part, of the common channel **12**, which is formed in the plate **128**. Since the common channel **12** has the protruded part **212A**, the first communicating channel **217A2** and the common channel **12** are less likely to interfere with each other, even in a case where the first communicating channel **217A2** is formed in the channel member **21**. Owing to this, the first communicating channel **217A2** which extends from the upper end of the first up-down channel **217A1** to the first pressure chamber **16A** along the third direction **D7** is easily formed in the channel member **21**.

[0087] As a second modification, the first individual channel **13A** may include a first communicating channel **317A2** having a horizontal part **317A2A** depicted in FIG. **8**, instead of the horizontal part **17A2A** described above; and the second individual channel **13B** may include a communicating channel **317B** depicted in FIG. **8A**, instead of the communicating channel **17B** described above. The communicating channel **317B** corresponds to the “second up-down channel” of the present disclosure.

[0088] The horizontal part **317A2A** in the second modification is connected to the upper end of the first up-down channel **17A1** as depicted in FIG. **8A**. The horizontal part **317A2A** is a hole formed in the plate **125**, and extends from the upper end of the first up-down channel **17A1** in a fourth direction **D4** parallel to the second direction **D2**. Further, as depicted in FIG. **8B**, the horizontal part **317A2A** has a throttle part **317A2C** which throttles the flow amount of the liquid. The throttle part **317A2C** is formed in a central part in the fourth direction **D4** of the horizontal part **317A2A**, and the throttle part **317A2C** has a channel width (length in the direction orthogonal to the fourth direction **D4**) which is smaller than the channel width of the first up-down channel **17A1**. In the second modification, although both ends in the fourth direction **D4** of the horizontal part **317A2A** have a greater channel width than the channel width of the throttle part **317A2C**, the horizontal part **317A2A** may have a channel width which is constant over the entire length of the horizontal part **317A2A**.

[0089] As depicted in FIG. **8A**, the communicating channel **317B** extends upward from the second

nozzle **15B** in the up-down direction **D3**. The communicating channel **317B** is constructed of nine holes formed, respectively, in the nine plates **122** to **130** and connected to one another, and has a diameter greater than the diameter of the second nozzle **15B**. The hole formed in the plate **122** has a hole diameter smaller than the diameter of each of the holes formed in the other plates **123** to **130**. In other words, the communicating channel **317B** also has a throttle part **317B2C** in the upper part thereof which throttles the flow amount of the liquid. The channel width (hole diameter) of the throttle part **317B2C** is smaller than the channel width of another part, of the communicating channel **317B**, which is other than the upper part. The channel shape and size of the communicating channel **317B** are adjusted so that the communicating channel **317B** has the channel resistance and inertance which are same as the channel resistance and inertance of the communicating channel **17A** including the horizontal part **317A2A**. Owing to this, the effects similar to the effect described above can be obtained.

[0090] The first communicating channel **317A2** and the communicating channel **317B** have the throttle part **317A2C** and the throttle part **317B2C**, respectively. Such a simple configuration is capable of reducing higher-order frequency component(s) in the individual channel **13** generated by the application of pressure to the ink in the pressure chamber **16** in a case where the ink is (to be) ejected from the nozzle **15**. By reducing the higher-order frequency component(s), the pressure transmitted to the nozzle **15** is also stabilized, allowing a predetermined amount of ink to be ejected from the nozzle **15** in a stable manner.

## Second Embodiment

### Head **401**

[0091] A head **401** according to a second embodiment of the present disclosure includes a channel member **421** and an actuator member **422**, as depicted in FIG. **9**. Both the channel member **421** and the actuator member **422** have a rectangular shape in which the length thereof in the first direction **D1** is longer than the length thereof in the second direction **D2** in the plane orthogonal to the up-down direction **D3**.

[0092] As depicted in FIG. **9**, three supply ports **4111** and three return ports **4112** are open in the upper surface (surface) **421A** of the channel member **421**. The three supply ports **4111** are located at one end in the first direction **D1** of the channel member **421**. The three return ports **4112** are located at the other end in the first direction **D1** of the channel member **421**. Each of the three supply ports **4111** and the three return ports **4112** is connected to an ink tank via a tube. The channel member **421** has three common channels **412**, and a plurality of individual channels **413**.

[0093] The three common channels **412** are disposed side by side in the second direction **D2** and each extend in the first direction **D1**. Each of the three supply ports **4111** is connected to one end in the first direction **D1** of one of the three common channels **412**. Each of the three return ports **4112** is connected to the other end in the first direction **D1** of one of the three common channels **412**. Each of the three common channel **412** communicates with the ink tank via a corresponding supply port **4111** among the three supply ports **4111** and a corresponding return port **4112** among the three return ports **4112**, and communicates with the plurality of individual channels **413**.

[0094] As depicted in FIGS. **10** to **12B**, each of the plurality of individual channels **413** includes a nozzle **415**, a pressure chamber **416**, a communicating channel **417**, a connecting channel **418**, and an up-down channel **419**. One end of the up-down channel **419** communicates with the nozzle **415**, and the other end of the up-down channel **419** communicates with the communicating channel **417**. One end of the communicating channel **417** communicates with the up-down channel **419**, and the other end of the communicating channel **417** communicates with the pressure chamber **416**. One end of the connecting channel **418** communicates with the common channel **412**, and the other end of the connecting channel **418** communicates with the pressure chamber **416**.

[0095] As depicted in FIGS. **11A** and **11B**, the channel member **421** includes eleven plates **4121** to **4131**. Note that the channel member **421** may be constructed of eleven or more plates or ten or less plates. The pressure chamber **416** is formed as a plurality of pressure chambers **416** in the plate

**4121** which is the uppermost layer among the eleven plates **4121** to **4131**, and the nozzle **415** is formed as a plurality of nozzles **415** in the plate **4131** which is the lowermost layer among the eleven plates **4121** to **4131**.

[0096] The plurality of pressure chambers **416** are open in the upper surface (upper surface **421A**) of the plate **4121**, and the plurality of nozzles **415** are open in the lower surface of the plate **4131**. The opening of each of the nozzles **415** is circular, and the opening of each of the pressure chambers **416** is substantially rectangular which is elongated in the second direction **D2**. In other words, the length (width) in the first direction **D1** of the pressure chamber **416** is shorter than the length in the second direction **D2** of the pressure chamber **416**. The nozzle **415** has a shape tapered downward, as depicted in FIGS. **11A** and **11B** and FIGS. **12A** and **12B**. Each of the three common channels **412** is constructed of four holes formed, respectively, in the four plates **4127** to **4130** and connected to one another.

[0097] As depicted in FIG. **9**, the plurality of individual channels **413** are aligned in the first direction **D1** and construct 12 individual channel rows **414**. These 12 individual channel rows **414** are disposed side by side in the second direction **D2**. In each of the 12 individual channel rows **414**, the plurality of nozzles **415** are aligned at a predetermined pitch **P** in the first direction **D1**. Further, all of the nozzles **415** are located at mutually different positions in the first direction **D1**. With this, in a case where the recording resolution at the pitch **P** is 100 dpi, a recording resolution of 1200 dpi is realized by all the nozzles **415**.

[0098] Four rows of the 12 individual channel rows **414** correspond to one of the three common channels **412**. For example, as depicted in FIGS. **10** to **12B**, the four individual channels rows **414** which are a first individual channel row **414A** to a fourth individual channel row **414D** include first individual channels **413A**, second individual channels **413B**, third individual channels **413C** and fourth individual channels **413D**, respectively. The first individual channels **413A**, the second individual channels **413B**, the third individual channels **413C** and the fourth individual channels **413D** communicate with the same common channel **412** among the three common channels **412**. Therefore, it can be considered that four rows of the 12 individual channel rows **414** correspond to one of the three common channels **412**.

[0099] As depicted in FIG. **10**, among the first individual channel rows **414A** to the fourth individual channel rows **414D** corresponding to each of the three common channels **412**, each of first nozzles **415A** included in the first individual channel row **414A** and each of second nozzles **415B** included in the second individual channel row **414B** are located on one side in the second direction **D2** (the right side in FIG. **10**) with respect to a corresponding common channels **412** among the three common channels **412**. Among the first individual channel rows **414A** to fourth individual channel rows **414D** corresponding to each of the three common channels **412**, each of third nozzles **415C** included in the third individual channel row **414C** and each of fourth nozzles **415D** included in the fourth individual channel row **414D** are located on the other side in the second direction **D2** (the left side in FIG. **10**) with respect to the corresponding common channels **412** among the three common channels **412**.

[0100] All the plurality of individual channels **413** in the second embodiment have the same channel structure including the channel shape and size. A first individual channel **413A** and a second individual channel **413B** which are adjacent in the second direction **D2** are located to be separated from each other in the second direction **D2** so that the first nozzle **415A** overlaps with the second pressure chamber **416B** in the up-down direction **D3**, and that the first nozzle **415A** and the second nozzle **415B** are separated from each other by a distance corresponding to 1/12 of the pitch **P** in the first direction **D1**, as depicted in FIG. **10**. Similarly, a third individual channel **413C** and a fourth individual channel **413D** which are adjacent in the second direction **D2** are located to be separated from each other in the second direction **D2** so that the third nozzle **415C** overlaps with the fourth pressure chamber **416D** in the up-down direction **D3**, and that the third nozzle **415C** and the fourth nozzle **415D** are separated from each other by a distance corresponding to 1/12 of the

pitch P in the first direction D1. Further, a second individual channel 413B and a fourth individual channel 413D which are adjacent in the second direction D2 are located so that the second nozzle 415B overlaps with the fourth pressure chamber 416D in the up-down direction D3 and that the fourth nozzle 415D overlaps with the second pressure chamber 416B in the up-down direction D3. [0101] The first individual channel 413A and the third individual channel 413C are located point-symmetrically with respect to a middle point of a line segment connecting the first nozzle 415A and the third nozzle 415C in the plane orthogonal to the up-down direction D3. The second individual channel 413B and the fourth individual channel 413D are located point-symmetrically with respect to a middle point of a line segment connecting the second nozzle 415B and the fourth nozzle 415D in the plane orthogonal to the up-down direction D3.

[0102] The first individual channel 413A and the fourth individual channel 413D are also located point-symmetrically with respect to a middle point of a line segment connecting the first nozzle 415A and the fourth nozzle 415D in the plane orthogonal to the up-down direction D3. The second individual channel 413B and the third individual channel 413C are also located point-symmetrically with respect to a middle point of a line segment connecting the second nozzle 415B and the third nozzle 415C in the plane orthogonal to the up-down direction D3.

[0103] Further, the first nozzle 415A, the second nozzle 415B, the third nozzle 415C and the fourth nozzle 415D as the four nozzles 415 are located between two common channels 412 which are included in the three common channels 412 and which are adjacent in the second direction D2, as depicted in FIG. 13. The first nozzles 415A to the fourth nozzle 415D are located in the order of the first nozzle 415A, the fourth nozzle 415D, the second nozzle 415B, and the third nozzle 415C from the left in FIG. 13. The first nozzle 415A and the fourth nozzle 415D are separated by a distance corresponding to  $\frac{1}{6}$  of the pitch P in the first direction D1. The second nozzle 415B and the third nozzle 415C are also separated by the distance corresponding to  $\frac{1}{6}$  of the pitch P in the first direction D1. In other words, the first nozzles 415A to the fourth nozzle 415D are located in a staggered manner in the second direction D2.

[0104] In other words, among the first nozzles 415A to the fourth nozzle 415D, one nozzle 415 or two nozzles 415 are not located, in the plane orthogonal to the up-down direction D3, on a straight line passing through two nozzles 415 which are adjacent in the second direction D2 and which are other than the one nozzle 415 or the two nozzles 415. Namely, as depicted in FIG. 13, the second nozzle 415B and the third nozzle 415C are not located on a straight line L1 passing through the first nozzle 415A and the fourth nozzle 415D, the first nozzle 415A and the third nozzle 415C are not located on a straight line L2 passing through the fourth nozzle 415D and the second nozzle 415B, and the first nozzle 415A and the fourth nozzle 415D are not located on a straight line L3 passing through the second nozzle 415B and the third nozzle 415C.

[0105] By locating the first nozzle 415A to the fourth nozzle 415D in this manner, projection points NA to ND (projection points indicated by hollow circles in FIG. 14A) which are projected, respectively, from the first nozzle 415A to the fourth nozzle 415D along the second direction D2 and onto a straight line L parallel to the first direction D1 are located at equal distances in the first direction D1, as depicted in FIG. 14A. In another head wherein the first nozzle 415A to the fourth nozzle 415D are located to be separated from one another at a distance corresponding to  $\frac{1}{12}$  of the pitch P to one side in the first direction DI (downward in FIG. 14B) further toward one side in the second direction D2 (rightward in FIG. 14B) as depicted in FIG. 14B, projection points NA to ND (projection points indicated by hollow circles in FIG. 14B) which are projected, respectively, from the first nozzle 415A to the fourth nozzle 415D along the second direction D2 and onto a straight line L in a manner similar to the manner in the above-described second embodiment are located at equal distances in the first direction D1.

[0106] However, in the manner in which the first nozzle 415A to the fourth nozzle 415D are located in the another head as depicted in FIG. 14B and in a case where the another head is disposed to be inclined at a predetermined angle  $\theta$  0 with respect to the first direction D1 in the

plane orthogonal to the up-down direction D3, each of the first nozzle **415A** to fourth nozzle **415D** moves to a position thereof indicated by a broken line. In this case, projection points NA to ND which are projected, respectively, from the first nozzle **415A** to the fourth nozzle **415D** onto the straight line L along the second direction D2 are located to overlap and merge substantially in one location.

[0107] On the other hand, in the manner in which the first nozzle **415A** to the fourth nozzle **415D** are located in the present embodiment and in a case where the head **1** is located to be inclined in the same manner as described above regarding the another head, each of the first nozzle **415A** to the fourth nozzle **415D** moves to a position thereof indicated by a broken line depicted in FIG. **14A**. In this case, although projection points NA and NC which correspond, respectively, to the two nozzles **415A** and **415C** among the projection points NA to ND which are projected, respectively, from the first nozzle **415A** to the fourth nozzle **415D** onto the straight line L along the second direction D2, are located to overlap and merge substantially in one location, projection points NB and ND which correspond, respectively, to the two nozzles **415B** and **415D** other than the nozzles **415A** and **415C** are located to be separated in the first direction D1. With this, since the first nozzle **415A** to the fourth nozzle **415D** are located in a staggered manner in the present embodiment as described above, even in a case where the channel member **421** is disposed to be inclined with respect to the first direction D1 in the plane orthogonal to the up-down direction D3, the resolution in the first direction D1 can be prevented from lowering.

[0108] The detailed configuration of the first individual channel **413A** will be described below.

[0109] As depicted in FIG. **11A**, each of the first individual channels **413A** includes a first nozzle **415A**, a first pressure chamber **416A**, a communicating channel **417A**, a connecting channel **418A**, and an up-down channel **419A**. As depicted in FIG. **10**, the first pressure chamber **416A** overlaps, in the second direction D2, with the second pressure chamber **416B** included in the second individual channel **413B** adjacent to the first individual channel **413A**. The connecting channel **418A** corresponds to the “first connecting channel” of the present disclosure.

[0110] As depicted in FIG. **11A**, the up-down channel **419A** extends upward from the first nozzle **415A** along the up-down direction D3. As depicted in FIG. **10**, the up-down channel **419A** overlaps with the second pressure chamber **416B** included in the second individual channel **413B** in the up-down direction D3. Further, as depicted in FIG. **11A**, the up-down channel **419A** is constructed of five holes formed, respectively, in the five plates **4126** to **4130** which are connected to one another, and has a diameter greater than the diameter of the first nozzle **415A**. In the second direction D2, the first nozzle **415A** is located closer to the corresponding common channel **412** than the second nozzle **415B** included in the second individual channel **413B** is.

[0111] The communicating channel **417A** includes a first part **417A1** having one end connected to the first pressure chamber **416A**, and a second part **417A2** having one end connected to the other end of the first part **417A1** and the other end connected to the up-down channel **419A**. The first part **417A1** is constructed of three holes formed, respectively, in the three plates **4122** to **4124** and connected to one another. The three holes constructing the first part **417A1** are formed so that the diameters thereof become smaller from the plate **4122** further toward the plate **4124**. In other words, the cross-sectional area of the first part **417A1** becomes smaller in a stepped manner further downward from the first pressure chamber **416A**. Further, as depicted in FIG. **11A** and FIG. **13**, the first part **417A1** extends further downward from the first pressure chamber **416A** in a fifth direction D5 crossing the first direction D1 and the second direction D2.

[0112] As depicted in FIG. **11A** and FIG. **13**, the second part **417A2** is a hole formed in the plate **4125**, and extends, in the plane orthogonal to the up-down direction D3, in a third direction D4 crossing the first direction D1, the second direction D2, and the fifth direction D5. That is, the communicating channel **417A** has a bent shape in the plane orthogonal to the up-down direction D3, as depicted in FIG. **13**. The second part **417A2** in the second embodiment has a length orthogonal to the third direction D4 which is smaller than the hole diameter of the up-down

channel **419A**. The second part **417A2** corresponds to the “first extending part” of the present disclosure.

[0113] As depicted in FIG. **11A**, the connecting channel **418A** has a first part **418A1** and a second part **418A2**. The first part **418A1** is constructed of three holes formed, respectively, in the three plates **4124** to **4126** and connected to one another, and is connected to an upper end of the common channel **412**.

[0114] The second part **418A2** has a horizontal part **418A2A** connected to the upper end of the first part **418A1** and a vertical part **418A2B** connected to the first pressure chamber **416A**. The horizontal part **418A2A** is a hole formed in the plate **4123**, and extends from the upper end of the first part **418A1** toward the first pressure chamber **416A** along a fourth direction **D6** (see FIG. **13**) which is parallel to the second direction **D2**. Further, as depicted in FIG. **13**, the horizontal part **418A2A** has a throttle part **418A2C** which throttles a flow amount of the liquid. The throttle part **418A2C** is formed in a central part in the fourth direction **D6** of the horizontal part **418A2A**, and has a channel width (length in the direction orthogonal to the fourth direction **D6**) smaller than a channel width of the first part **418A1**. In the present embodiment, although both ends in the fourth direction **D6** of the horizontal part **418A2A** have a channel width greater than the channel width of the throttle part **418A2C**, the horizontal part **418A2A** may have a channel width which is constant over the entirety of the length of the horizontal part **418A2A**. The horizontal part **418A2A** corresponds to the “second extending part” of the present disclosure.

[0115] As depicted in FIG. **11A**, the vertical part **418A2B** is a hole formed in the plate **4122**, and extends upward toward the first pressure chamber **416A**, from an end, of the horizontal part **418A2A**, which is close to the first pressure chamber **416A**. In this manner, the second part **418A2** is constructed of the two holes formed, respectively, in the two plates **4122**, **4123** and connected to each other.

[0116] The second individual channel **413B**, the third individual channel **413C** and the fourth individual channel **413D** have a configuration similar to the configuration of the first individual channel **413A** as described above, and thus the configuration of each of the second individual channel **413B**, the third individual channel **413C** and the fourth individual channel **413D** will not be described in detail. As depicted in FIG. **11B**, the second individual channel **413B** includes a second nozzle **415B**, a second pressure chamber **416B**, a communicating channel **417B**, a first part **417B1**, a second part **417B2**, a connecting channel **418B**, a first part **418B1**, a second part **418B2**, a horizontal part **418B2A**, a vertical part **418B2B**, a throttle part **418B2C**, and an up-down channel **419B**. The connecting channel **418B** corresponds to the “second connecting channel” of the present disclosure.

[0117] As depicted in FIG. **12A**, the third individual channel **413C** includes a third nozzle **415C**, a third pressure chamber **416C**, a communicating channel **417C**, a first part **417C1**, a second part **417C2**, a connecting channel **418C**, a first part **418C1**, a second part **418C2**, a horizontal part **418C2A**, a vertical part **418C2B**, a throttle part **418C2C**, and an up-down channel **419C**. As depicted in FIG. **12B**, the fourth individual channel **413D** includes a fourth nozzle **415D**, a fourth pressure chamber **416D**, a communicating channel **417D**, a first part **417D1**, a second part **417D2**, a connecting channel **418D**, a first part **418D1**, a second part **418D2**, a horizontal part **418D2A**, a vertical part **418D2B**, a throttle part **418D2C**, and an up-down channel **419D**.

[0118] As depicted in FIG. **13**, the third pressure chamber **416C** overlaps with the fourth pressure chamber **416D** in the second direction **D2**. More specifically, the first pressure chamber **416A** to the fourth pressure chamber **416D** included, respectively, in the first individual channel **413A** to the fourth individual channels **413D** overlap with one another in the second direction **D2**. In the second direction **D2**, the third nozzle **415C** is located closer to the corresponding common channel **412** than the fourth nozzle **415D** is.

[0119] The second part **417B2** of the communicating channel **417B** included in the second individual channel row **414B** and the second part **417D2** of the communicating channel **417D**



included in the fourth individual channel row **414D** overlap each other in the first direction **D1** and are separated from each other in the first direction **D1**. Owing to this, the communicating channels **417B** and **417D** which are adjacent can be separated by a desired tolerance or more while locating the nozzles **415** at a high density. Accordingly, the communicating channels **417B** and **417D** can be prevented from communicating with each other due to an error during the manufacture of the channel member **421**.

[0120] The ink in the ink tank is supplied to each of the three common channels **412** via one of the three supply ports **4111** corresponding thereto by driving the pump **10** depicted in FIG. 2 under the control of the controller **5**, and then the ink is distributed from the three common channels **412** to the plurality of individual channels **413**.

[0121] In each of the plurality of individual channels **413**, the volume of the pressure chamber **416** is reduced by driving an actuator part **435** which will be described later, and pressure is applied to the ink in the pressure chamber **416**, thereby causing the ink to pass through the communicating channel **417** and to be ejected as an ink droplet from the nozzle **415**.

[0122] The ink supplied from each of the three supply ports **4111** moves in one of the three common channels **412** corresponding thereto from the one end to the other end in the first direction **D1**, and reaches one of the three return ports **4112** corresponding thereto. The ink which reaches each of the three return ports **4112** is returned to the ink tank via the tube.

[0123] As depicted in FIGS. 9, 11A, 11B, 12A and 12B, the actuator member **422** is fixed to the upper surface **421A** of the channel member **421**. The actuator member **422** includes a vibration plate **431** made of metal, a piezoelectric layer **432**, and a plurality of individual electrodes **433**.

[0124] Parts, of the actuator member **422**, each of which overlaps with one of the plurality of pressure chambers **416** corresponding thereto in the up-down direction **D3** function as actuator parts **435**. Each of the actuator parts **435** is independently deformable in accordance with the potential applied to one of the plurality of individual electrodes **433** corresponding thereto.

[0125] Each of the actuator parts **435** is a thin-film piezoelectric element. The thin-film piezoelectric element is a so-called micro electro mechanical system (MEMS). The actuator parts **435** is formed by sequentially depositing a thin film which becomes the piezoelectric layer **432** and a thin film which becomes the plurality of individual electrodes **433**, on the upper surface of the vibration plate **431**.

[0126] The vibration plate **431** is disposed on the upper surface **421A** of the channel member **421** so as to cover the plurality of pressure chambers **416**. The piezoelectric layer **432** is disposed on the upper surface of the vibration plate **431**. Each of the plurality of individual electrodes **433** is disposed on the upper surface of the piezoelectric layer **432** so as to overlap with each of the pressure chambers **416** corresponding thereto in an up-down direction **D3**.

[0127] The vibration plate **431** and the plurality of individual electrodes **433** are electrically connected to the driver IC **6**. The driver IC **6** maintains the potential of the vibration plate **431** at the ground potential, whereas the driver IC **6** changes the potential of each of the plurality of individual electrodes **433**. The vibration plate **431** functions as a common electrode which is common to the plurality of actuator parts **435**.

[0128] The driver IC **6** generates a drive pulse signal based on a control signal from the controller **5**, and supplies the drive pulse signal to each of the plurality of individual electrodes **433**. The drive pulse signal changes the potential of each of the plurality of individual electrodes **433** between a predetermined driving potential and the ground potential. In this manner, the actuator part **435** is driven, and the pressure is applied to the ink in the pressure chamber **416**, causing an ink droplet to pass through the communicating channel **417** and to be ejected from the nozzle **415**.

[0129] As described above, according to the head **401** of the second embodiment, no individual channels **413** having different channel shapes are not included in the plurality of individual channels **413** connected to the same common channel **412**. Therefore, even in a case where the plurality of nozzles **415** are located at high density, the liquid ejection characteristic is less likely to

vary among the plurality of nozzles **415**.

[0130] The first nozzle **415A** overlaps with the second pressure chamber **416B** in the up-down direction **D3**. Owing to this, the channel member **421** can be prevented from becoming large-sized in the second direction **D2**.

[0131] The third nozzle **415C** overlaps with the fourth pressure chamber **416D** in the up-down direction **D3**. Owing to this, the channel member **421** can be prevented from becoming large-sized in the second direction **D2** while locating the plurality of nozzles **415** at high density.

[0132] The cross-sectional area of each of the first parts **417A1** to **417D1** of the communicating channel **417** becomes smaller in a stepped manner further downward from the pressure chamber **416**. In such a presumed case where the upper end part of each of the first parts **417A1** to **417D1** is configured to have a smaller cross-sectional area than the cross-sectional area of the lower end part thereof, the pressure applied to the ink in the pressure chamber **416** is thereby rapidly attenuated and is less likely to be transmitted to the ink in the nozzle **415**. In the present (or “second”?) embodiment, however, since the cross-sectional area of each of the first parts **417A1** to **417D1** becomes smaller in the stepped manner, the pressure applied to the ink in the pressure chamber **416** in a case where the ink is (to be) ejected from the nozzle **415** is easily transmitted to the ink in the nozzle **415**. Owing to this, the ink can be easily ejected from the nozzle **415**.

[0133] The second parts **417A2** to **417D2** of the communicating channel **417** and the horizontal parts **418A2A** to **418D2A** of the connecting channel **418** are located at mutually different heights. This prevents the second parts **417A2** to **417D2** and the horizontal parts **418A2A** to **418D2A** from interfering with one another in a case where the plurality of individual channels **413** are located adjacent to one another. Owing to this, the plurality of individual channels **413** can be located close to one another, thereby reducing the size of the channel member **413** in the direction orthogonal to the up-down direction **D3**.

[0134] The channel member **421** is constructed of the plurality of stacked plates **4121** to **4131**. Owing to this, each of the pressure chambers **416**, the nozzles **415**, etc., in the plurality of individual channels **413** can be formed in the same plate.

[0135] The head **1** includes the actuator member **422** having the plurality of actuator parts **435** each corresponding to one of the plurality of pressure chambers **416**. In this case, the head **401** is less susceptible to the influence of non-uniformity during the manufacture of the actuator member **422** itself than in a case where the head **401** includes a plurality of actuator members each of which has the actuator part **435** corresponding to one of the pressure chambers **416**. Therefore, in the head **401** of the second embodiment, the ink ejection characteristic is less likely to vary among the plurality of nozzles **415**.

[0136] All the plurality of nozzles **415** are located to be shifted from one another in the first direction **D1**, thereby improving the resolution in the first direction **D1**.

[0137] The connecting position at which the connecting channel **418B** connects to the common channel **412** is separated, in the second direction **D2**, from the connecting position at which the connecting channel **418A** connects to the common channel **412**. In a presumed case where these two connecting positions are not separated in the second direction **D2** but are located side by side in the first direction **D1**, the ink at the same position in the second direction **D2** of the common channel **412** flows to the individual channel **413**. Due to this, ink each at positions separated from the connecting position with respect to the common channel **412** in the second direction **D2** stagnates without flowing to the individual channel **413**. However, in the second embodiment, since these two connecting positions are separated in the second direction **D2**, the ink flows from at least two positions in the second direction **D2** of the common channel **412** to the individual channel **413**. Therefore, the stagnation of the ink is less likely to occur in the common channel **412**.

[0138] While the invention has been described in conjunction with various example structures outlined above and illustrated in the figures, various alternatives, modifications, variations, improvements, and/or substantial equivalents, whether known or that may be presently unforeseen,

may become apparent to those having at least ordinary skill in the art. Accordingly, the example embodiments of the disclosure, as set forth above, are intended to be illustrative of the invention, and not limiting the invention. Various changes may be made without departing from the spirit and scope of the disclosure. Therefore, the disclosure is intended to embrace all known or later developed alternatives, modifications, variations, improvements, and/or substantial equivalents. Some specific examples of potential alternatives, modifications, or variations in the described invention are provided below:

[0139] Although the present disclosure have been described above regarding the first and second embodiments and the modifications, the present disclosure is not limited to the above-described first and second embodiments and the modifications. Various changes can be made within the scope of the claims.

[0140] In the above-described first and second embodiments, the nozzles included, respectively, in the two individual channel rows **14**, **412** corresponding to each of the common channels **12**, **412** are located on each of the both sides in the second direction D2 with respect to the corresponding common channel **12**, **412**. However, the nozzles included, respectively, in the two individual channel rows may be located only on one side in the second direction D2 with respect to the corresponding common channel **12**, **412**.

[0141] Further, a plurality of actuator members each having the actuator part **35**, **435** may be disposed on the upper surface **21A**, **421A** of the channel member **21**, **421**, instead of the actuator member **22**, **422**.

[0142] Furthermore, the first individual channel **13A** and the second individual channel **13B** may have mutually different channel resistances or inertances. Moreover, the first individual channel **13A** and the third individual channel **13C** may have mutually different channel structures, including the channel shape and size. Further, the second individual channel **13B** and the fourth individual channel **13D** may also have mutually different channel structures, including the channel shape and size.

[0143] The connecting channels **18A** and **18B** may have mutually different channel lengths and/or channel widths. Further, the connecting position at which the connecting channel **18A** connects to the common channel **12** and the connecting position at which the connecting channel **8B** connects to the common channel **12** may be located side by side in the first direction D1.

[0144] The horizontal part **18B2A** of the connecting channel **18B** and the horizontal part **17A2A** of the first communicating channel **17A2** may be located at the same height. The horizontal part **18C2A** of the connecting channel **18C** and the horizontal part **18A2A** of the connecting channel **18A** do not have to be disposed side by side in the first direction D1.

[0145] In the above-described embodiments, although the electrodes constructing the actuator part **35** have a two-layered structure including the individual electrodes and the common electrode, the electrodes constructing the actuator part **35** may have a three-layered structure. For example, the three-layered structure means a configuration including a driving electrode to which a high potential and a low potential are selectively applied, a high potential electrode which is maintained at the high potential and a low potential electrode which is maintained at the low potential.

[0146] Further, the nozzle **415** included in one individual channel **413** does not have to overlap with the pressure chamber **416** included in another individual channel **413** in the up-down direction D3.

[0147] Furthermore, the second part **417B2** of the communicating channel **417B** and the second part **417D2** of the communicating channel **417D** do not have to overlap each other in the first direction D1. Moreover, the cross-sectional area of each of the first parts **417A1** to **417D1** of the communicating channel **417** may be the same or may increase in a stepped manner further downward from the pressure chamber **416**.

[0148] The second parts **417A2** to **417D2** of the communicating channel **417** and the horizontal parts **418A2A** to **418D2A** of the connecting channel **418** may be located at the same height.

[0149] Among all the plurality of nozzles **415**, nozzles **415** as a part of the plurality of nozzles **415** may be located at the same position in the first direction **D1**. As depicted in FIG. **14B**, the first nozzle **415A** to the fourth nozzle **415D** may be located to be separated from one another at the distance corresponding to  $1/12$  of the pitch **P** to the one side in the first direction **D1** (downward in FIG. **14B**) further toward the one side in the second direction **D2** (rightward in FIG. **14B**).

[0150] In the second direction **D2**, the connection position at which the connecting channel **418B** connects to the common channel **412** may be the same position as the connecting position at which the connecting channel **418A** connects to the common channel **412**.

[0151] The type of the liquid ejecting head is not limited to the line system, but may also be the serial system.

[0152] The object to which the liquid droplet is to be ejected is not limited to the sheet. For example, the object to which the liquid droplet is to be ejected may be cloth, a substrate, or plastic.

[0153] The liquid droplet ejected from the nozzle is not limited to the ink droplet. For example, the liquid droplet may be a droplet of a treatment liquid which agglutinates or precipitates a component in an ink.

[0154] The present disclosure is not limited to being applicable to printers, and is applicable also to facsimiles, copying machines, and multi-function peripherals. Further, the present disclosure is applicable also to a liquid droplet ejecting head for any usage other than the image recording. For example, the present disclosure is applicable to a liquid ejecting head which ejects a conductive liquid to a substrate so as to form a conductive pattern.

## Claims

**1.** A liquid ejecting head comprising a channel member including: a common channel extending in a first direction orthogonal to an up-down direction; and a plurality of individual channels each including a nozzle and a pressure chamber communicating with the nozzle, each of the plurality of individual channels communicating with the common channel, wherein a plurality of individual channel rows each including the plurality of individual channels aligned in the first direction are disposed side by side in a second direction orthogonal to the up-down direction and the first direction, a first pressure chamber and a second pressure chamber overlap in the second direction, the first pressure chamber being included in a first individual channel row among the plurality of individual channel rows and the second pressure chamber being included in a second individual channel row adjacent to the first individual channel row among the plurality of individual channel rows, a first nozzle included in the first individual channel row and a second nozzle included in the second individual channel row are located on one side in the second direction with respect to the common channel, and the first nozzle is located closer to the common channel than the second nozzle is in the second direction, and the first nozzle overlaps with the second pressure chamber in the up-down direction.

**2.** The liquid ejecting head according to claim 1, wherein a plurality of pressure chambers which includes the pressure chamber and each of which is included in one of the plurality of individual channels are open in a surface of the channel member, and the liquid ejecting head further comprising an actuator member disposed in the surface and having a plurality of actuator parts each overlapping with one of the plurality of pressure chambers in the up-down direction.

**3.** The liquid ejecting head according to claim 1, wherein a first individual channel included in the first individual channel row includes a first up-down channel extending from the first nozzle in the up-down direction, and a first communicating channel communicating with the first pressure chamber and the first up-down channel, and the first communicating channel extends, without bending, at a location below the second pressure chamber in a third direction crossing the second direction and the up-down direction.

**4.** The liquid ejecting head according to claim 3, wherein the first pressure chamber and the first

communicating channel overlap with the common channel in the up-down direction, and an upper surface of the common channel includes a protruded part protruding into the common channel in an area overlapping with the first communicating channel.

**5.** The liquid ejecting head according to claim 1, wherein a channel resistance of a first individual channel included in the first individual channel row is equal to a channel resistance of a second individual channel included in the second individual channel row.

**6.** The liquid ejecting head according to claim 1, wherein an inertance of a first individual channel included in the first individual channel row is equal to an inertance of a second individual channel included in the second individual channel row

**7.** The liquid ejecting head according to claim 5, wherein the first individual channel includes a first connecting channel connecting the first pressure chamber and the common channel, the second individual channel includes a second connecting channel connecting the second pressure chamber and the common channel, and a length and a width of the first connecting channel are equal to a length and a width of the second connecting channel, respectively.

**8.** The liquid ejecting head according to claim 5, wherein a third pressure chamber and a fourth pressure chamber overlap in the second direction, the third pressure chamber being included in a third individual channel row among the plurality of individual channel rows and the fourth pressure chamber being included in a fourth individual channel row adjacent to the third individual channel row among the plurality of individual channel rows, a third nozzle included in the third individual channel row and a fourth nozzle included in the fourth individual channel row are located on the other side in the second direction with respect to the common channel in the second direction, the third nozzle is located closer to the common channel than the fourth nozzle is in the second direction, and overlaps with the fourth pressure chamber in the up-down direction, a channel configuration of the first individual channel is equal to a channel configuration of a third individual channel included in the third individual channel row, and a channel configuration of the second individual channel is equal to a channel configuration of a fourth individual channel included in the fourth individual channel row.

**9.** The liquid ejecting head according to claim 5, wherein the first individual channel includes a first up-down channel extending from the first nozzle in the up-down direction, and a first communicating channel communicating with the first pressure chamber and the first up-down channel, a second individual channel includes a second up-down channel communicating with the second pressure chamber and the second nozzle, extending from the second nozzle in the up-down direction, and having a length in the up-down direction longer than a length in the up-down direction of the first up-down channel, and the first communicating channel has a part extending in a fourth direction orthogonal to the up-down direction.

**10.** The liquid ejecting head according to claim 1, wherein a first individual channel included in the first individual channel row includes a first up-down channel extending from the first nozzle in the up-down direction, and a first communicating channel communicating with the first pressure chamber and the first up-down channel, a second individual channel included in the second individual channel row includes a second up-down channel communicating with the second pressure chamber and the second nozzle, extending from the second nozzle in the up-down direction, and having a length in the up-down direction longer than a length in the up-down direction of the first up-down channel, the first communicating channel has a channel width smaller than a channel width of the first up-down channel, and an upper part of the second up-down channel has a channel width smaller than a channel width of a remaining part, of the second up-down channel, other than the upper part.

**11.** The liquid ejecting head according to claim 1, wherein a first individual channel included in the first individual channel row includes a first connecting channel connecting the first pressure chamber and the common channel, a second individual channel included in the second individual channel row includes a second connecting channel connecting the second pressure chamber and the

common channel, and a connecting position at which the first connecting channel connects to the common channel and a connecting position at which the second connecting channel connects to the common channel are separated in the second direction.

**12.** The liquid ejecting head according to claim 1, wherein a first individual channel included in the first individual channel row includes a first up-down channel extending from the first nozzle in the up-down direction, and a first communicating channel communicating with the first pressure chamber and the first up-down channel, a second individual channel included in the second individual channel row includes a second connecting channel connecting the second pressure chamber and the common channel, the first communicating channel has a first extending part extending in a fourth direction orthogonal to the up-down direction, the second connecting channel has a second extending part extending in a fifth direction orthogonal to the up-down direction, and the first extending part and the second extending part are located at mutually different heights.

**13.** The liquid ejecting head according to claim 1, wherein a third pressure chamber and a fourth pressure chamber overlap in the second direction, the third pressure chamber being included in a third individual channel row among the plurality of individual channel rows and the fourth pressure chamber being included in a fourth individual channel row adjacent to the third individual channel row among the plurality of individual channel rows, a third nozzle included in the third individual channel row and a fourth nozzle included in the fourth individual channel row are located on the other side in the second direction with respect to the common channel, the third nozzle is located closer to the common channel than the fourth nozzle is in the second direction, and overlaps with the fourth pressure chamber in the up-down direction, a first individual channel included in the first individual channel row includes a first connecting channel connecting the first pressure chamber and the common channel, a third individual channel included in the third individual channel row includes a third connecting channel connecting the third pressure chamber and the common channel, the first connecting channel has a third extending part extending in a sixth direction orthogonal to the up-down direction, and the third connecting channel has a fourth extending part located side by side with the third extending part in the first direction and extending in the sixth direction.

**14.** The liquid ejecting head according to claim 1, wherein the plurality of individual channels have a same channel shape.

**15.** The liquid ejecting head according to claim 14, wherein the first nozzle is located closer to the common channel than the second nozzle is in the second direction, and overlaps with the second pressure chamber in the up-down direction orthogonal to the first direction and the second direction.

**16.** The liquid ejecting head according to claim 15, wherein the channel member includes: another common channel which extends in the first direction and is disposed side by side with the common channel in the second direction, and a plurality of individual channels each including a nozzle and a pressure chamber communicating with the nozzle, each of the plurality of individual channels communicating with the another common channel, a plurality of individual channel rows each including the plurality of individual channels communicating with the another common channel and aligned in the first direction are disposed side by side in the second direction, a third pressure chamber and fourth pressure chamber overlap in the second direction, the third pressure chamber being included in a third individual channel row among the plurality of individual channel rows communicating with the another common channel and the fourth pressure chamber being included in a fourth individual channel row adjacent to the third individual channel row among the plurality of individual channel rows, a third nozzle included in the third individual channel row and a fourth nozzle included in the fourth individual channel row are located on the other side in the second direction with respect to the another common channel, and are located between the common channel and the another common channel, all of the plurality of individual channels communicating with the another common channel have a same channel shape, the third nozzle is located closer to

the another common channel than the fourth nozzle is in the second direction, and overlaps with the fourth pressure chamber in the up-down direction, and the second nozzle overlaps with the fourth pressure chamber in the up-down direction, and the fourth nozzle overlaps with the second pressure chamber in the up-down direction.

**17.** The liquid ejecting head according to claim 16, wherein each of the plurality of individual channels includes an up-down channel extending from the nozzle in the up-down direction, and a communicating channel communicating with the pressure chamber and the up-down channel, each of the communicating channel included in the second individual channel row and the communicating channel included in the fourth individual channel row has a bent shape in a plane orthogonal to the up-down direction, and has a first part having one end connected to the pressure chamber corresponding thereto, and a second part having one end connected to the other end of the first part and the other end connected to the up-down channel, and the second part of the communicating channel included in the second individual channel row and the second part of the communicating channel included in the fourth individual channel row overlap each other in the first direction and are separated from each other in the first direction.

**18.** The liquid ejecting head according to claim 15, wherein each of the plurality of individual channels includes an up-down channel extending from the nozzle in the up-down direction, and a communicating channel communicating with the pressure chamber and the up-down channel, and the communicating channel has a part of which cross-sectional area decreases in a stepped manner further downward from the pressure chamber.

**19.** The liquid ejecting head according to claim 15, wherein each of the plurality of individual channels includes an up-down channel extending from the nozzle in the up-down direction, a communicating channel communicating with the pressure chamber and the up-down channel, and a connecting channel connecting the pressure chamber and the common channel, the communicating channel has a first extending part extending in a third direction orthogonal to the up-down direction, the connecting channel has a second extending part extending in a fourth direction orthogonal to the up-down direction, and the first extending part and the second extending part are located at mutually different heights.

**20.** The liquid ejecting head according to claim 14, wherein the channel member is constructed of a plurality of stacked plates.

**21.** The liquid ejecting head according to claim 14, wherein a plurality of pressure chambers which includes the pressure chamber and each of which is included in one of the plurality of individual channels are open in a surface of the channel member, and the liquid ejecting head further comprising an actuator member disposed on the surface and having a plurality of actuator parts each overlapping with one of the plurality of pressure chambers in the up-down direction.

**22.** The liquid ejecting head according to claim 16, wherein all of a plurality of nozzles which includes the nozzle and each of which is included in the plurality of individual channels communicating with the another channel member are located while being shifted from one another in the first direction.

**23.** The liquid ejecting head according to claim 22, wherein in a plane orthogonal to the up-down direction, one nozzle or two nozzles among the first nozzle to the fourth nozzles is or are not located on a straight line passing through two nozzles, among the first to fourth nozzles, which are adjacent in the second direction and other than the one nozzle or the two nozzles.

**24.** The liquid ejecting head according to claim 14, wherein a first individual channel included in the first individual channel row includes a first connecting channel connecting the first pressure chamber and the common channel, a second individual channel included in the second individual channel row includes a second connecting channel connecting the second pressure chamber and the common channel, and a connecting position at which the first connecting channel connects to the common channel and a connecting position at which the second connecting channel connects to the common channel are separated in the second direction.

